

Gdańsk University of Technology
Faculty of Electronics, Telecommunications and Informatics
Department of Multimedia Systems

Technical Documentation
Hardware – Fundus Camera Prototype
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**Smartphone-Based Fundus Camera:
Design and Assembly Guide**

Project Subject: Glaucoma detection using intelligent
analysis of fundus images

Team Members: Martyna Borkowska (Lead), Karolina Glaza,
Mateusz Dobry, Agnieszka Pawłowska,
Amila Amarasekara, Antoni Naczke

Supervisor: Prof. Andrzej Czyżewski, Ph.D., D.Sc., Eng.

Authors: Antoni Naczke

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1 Introduction

Fundus photography is one of the primary diagnostic tools used in ophthalmology for the detection and monitoring of retinal diseases, including glaucoma, diabetic retinopathy, and age-related macular degeneration. Commercial fundus cameras, however, are prohibitively expensive, often costing tens of thousands of euros, which limits their availability in resource-constrained clinical settings and developing countries.

This document presents the technical design and assembly documentation for a low-cost, smartphone-based fundus camera prototype developed as part of the Team 388 glaucoma detection project. The device leverages a smartphone's built-in camera and flashlight as the imaging sensor and illumination source, respectively, combined with a Volk 20D ophthalmic condensing lens and a fully 3D-printed mechanical assembly. The total material cost of the prototype is kept under 50, making it a viable option for screening applications.

1.1 Design Objectives

The fundus camera prototype was designed with the following key objectives:

- **Affordability** – utilize inexpensive, commercially available components and 3D-printed structural elements.
- **Portability** – the assembled device must be compact and lightweight enough for field use.
- **Adjustability** – the optical distance between the condensing lens and the patient's eye must be continuously adjustable.
- **Patient Comfort** – an ergonomic eye cup must be provided to stabilize the patient's head and reduce ambient light.
- **Image Quality** – the optical configuration must produce fundus images of sufficient quality for AI-based glaucoma screening.

2 Optical Principle

The device operates on the principle of **indirect ophthalmoscopy**. In this configuration, the Volk 20D condensing lens (focal length ≈ 50 mm) is held between the smartphone camera and the patient's eye. The lens produces a real, inverted, aerial image of the retina, which is then captured by the smartphone's camera sensor.

The smartphone's built-in flashlight (LED) serves as the illumination source. Light from the LED enters the patient's eye through the pupil, reflects off the retinal surface, exits through the pupil, and passes through the Volk 20D lens, forming a focused image at the camera sensor plane.

2.1 Key Optical Parameters

Parameter	Value
Condensing lens	Volk 20D
Lens focal length	≈ 50 mm
Working distance (lens to eye)	25 mm – 50 mm
Illumination source	Smartphone LED flash
Imaging sensor	Smartphone rear camera
Rail travel range	0 mm – 150 mm

Table 1: Summary of key optical parameters for the fundus camera prototype.

The adjustable rail system allows the operator to fine-tune the lens-to-eye distance in real time, accommodating patients with varying refractive errors and optimizing image focus.

3 Mechanical Design Overview

The mechanical structure of the fundus camera consists of three unique 3D-printed components (with the lens holder ring printed twice) connected by four metal rods that serve as a linear rail system. The design philosophy prioritizes simplicity of manufacture, ease of assembly, and adjustability during clinical use.

3.1 System Architecture

The complete assembly, from the smartphone end to the patient end, comprises the following elements arranged in sequence along the optical axis:

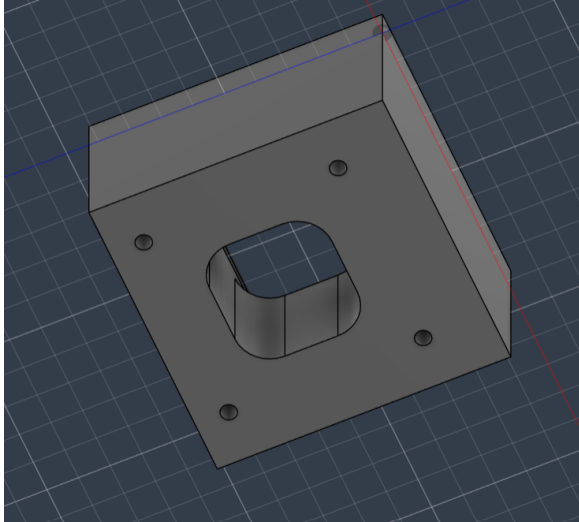
1. **Phone Adapter** (3D-printed) – mounts the smartphone in a dedicated cradle with a camera island cutout, and anchors the four guide rods.
2. **Four Metal Guide Rods** (15 cm each, 5 mm diameter) – serve as linear rails for the sliding lens holder assembly.
3. **Lens Holder Assembly** (3D-printed, 2 identical rings) – two identical rings that sandwich the Volk 20D lens between them. Bearings mounted in the rod tabs enable smooth sliding along the rods.
4. **Eye Cup Stabilizer** (3D-printed) – secures the distal ends of the rods and provides a mounting point for the rubber eye cup.

4 3D-Printed Components

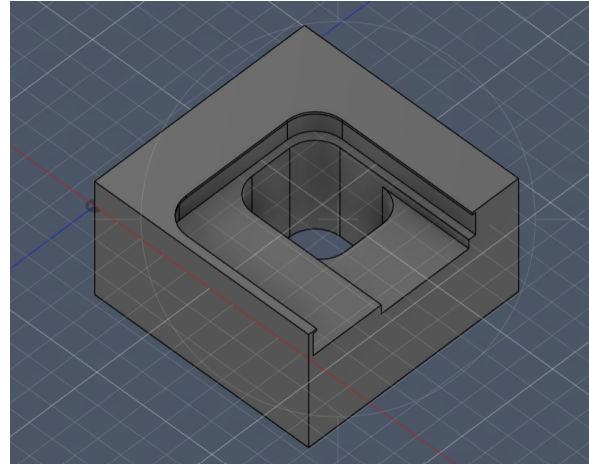
All structural components are manufactured using FDM (Fused Deposition Modeling) 3D printing. The recommended material is PLA (Polylactic Acid) due to its ease of printing, dimensional accuracy, and sufficient mechanical strength for this application. Standard FDM printing settings are suitable for all parts.

4.1 Component 1: Phone Adapter

The phone adapter is the proximal element of the assembly, directly interfacing with the smartphone. It is an L-shaped bracket consisting of a flat base plate with a perpendicular support wall. The component is designed with two distinct sides: the **rod-mounting side** (Figure 1a) facing the optical path, and the **phone cradle side** (Figure 1b) where the smartphone is seated.



(a) Rod-mounting side. The flat plate features a central camera aperture and four corner holes for the 5 mm guide rods.



(b) Phone cradle side. The recessed pocket accommodates the smartphone body, with a shaped cutout for the camera island (camera modules and LED flash).

Figure 1: CAD renderings of the Phone Adapter showing both sides of the component.

4.1.1 Design Features

- **Phone Cradle** – a recessed pocket on the rear face of the adapter, shaped to accept the smartphone body. The cradle holds the phone securely in place during imaging without the need for clamps or adhesives.
- **Camera Island Cutout** – a dedicated recess within the phone cradle that accommodates the smartphone's protruding camera island (including all camera modules and the LED flash), ensuring the phone sits flush against the adapter surface.
- **Central Camera Aperture** – a through-hole connecting the camera island cutout to the rod-mounting side, sized to align with the smartphone's primary rear camera. A raised cylindrical boss on the rod side helps center the optical path.
- **Four Rod Mounting Holes** – precision bores at the four corners of the base plate to accept the 5 mm metal guide rods via press-fit or friction-fit.
- **Perpendicular Support Wall** – an L-shaped wall rising from one edge of the base plate, providing a backstop against which the smartphone rests, preventing the phone from sliding during imaging.

4.1.2 Functional Role

The phone adapter ensures repeatable, stable positioning of the smartphone camera relative to the optical axis. The cradle design eliminates the need for separate phone clamps

and guarantees that the camera lens is centered on the optical path. The camera island cutout is critical for ensuring a flush fit, as modern smartphones feature protruding multi-camera modules that would otherwise prevent proper seating. The four corner holes establish the positions of the guide rods along which the lens holder assembly will travel.

4.2 Component 2: Lens Holder Ring ($\times 2$)

The lens holder ring is the central optical component carrier. **Two identical copies** of this ring are printed and used to enclose the Volk 20D condensing lens between them, gripping it firmly by the rim. The assembly slides along the four guide rods via linear bearings (LM5UU) integrated into one of the ring's tab sets.

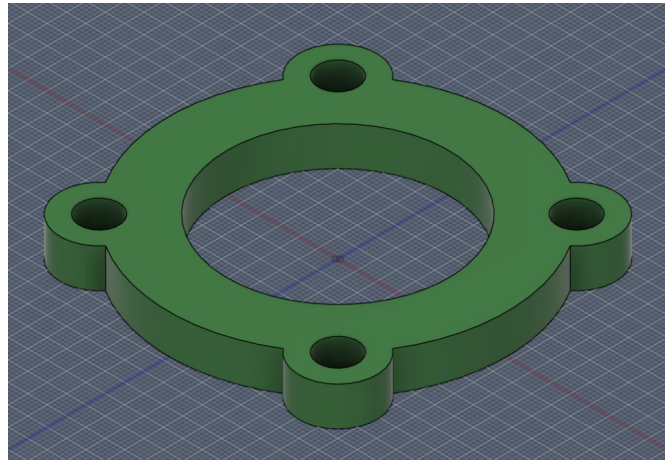


Figure 2: CAD rendering of the Lens Holder Ring. Two identical copies of this part are printed. The Volk 20D lens is clamped between them by the rim. The four protruding cylindrical tabs with through-holes accept the guide rods; one ring carries four LM5UU linear bearings (5 mm ID / 10 mm OD) for smooth sliding motion.

4.2.1 Design Features

- **Central Aperture** – a large circular opening sized to the clear aperture of the Volk 20D lens, allowing the optical path to pass through unobstructed.
- **Lens Clamping Face** – the inner face of each ring features a flat annular surface that contacts the rim of the Volk 20D lens. When the two rings are joined face-to-face, they grip the lens securely by its edge without contacting the optical surfaces.
- **Four Rod Tabs** – cylindrical protrusions at approximately 90° intervals around the ring's circumference. Each tab contains a precision bore sized to accept the 5 mm guide rods.
- **Bearing Seats** – on one of the two rings, the tab bores are dimensioned to accept LM5UU linear bearings (10 mm OD / 5 mm ID), providing low-friction guided motion along the rods. The second ring's tabs act as passive guides.

4.2.2 Assembly as a Pair

The two identical lens holder rings are assembled as follows:

1. Press-fit the four LM5UU linear bearings into the tab bores of the first ring.

2. Place the Volk 20D lens onto the clamping face of the bearing-equipped ring, centered within the aperture.
3. Place the second ring on top, face-to-face with the first, enclosing the lens between them.
4. Align the rod tab holes of both rings and bond them together using cyanoacrylate or epoxy adhesive.

4.2.3 Functional Role

The paired lens holder rings provide a secure, compact method of holding the Volk 20D lens. The rim-clamping approach ensures that no mechanical stress is applied to the optical surfaces, preserving image quality. The four linear bearings provide stable, wobble-free four-point contact with the guide rods, maintaining the lens perfectly perpendicular to the optical axis throughout its range of travel. By sliding the assembly along the rods, the operator can continuously adjust the lens-to-eye distance to optimize focus.

4.3 Component 3: Eye Cup Stabilizer

The eye cup stabilizer is the distal element of the assembly, positioned closest to the patient. It serves a dual purpose: securing the far ends of the four guide rods and providing a mounting point for the rubber eye cup.

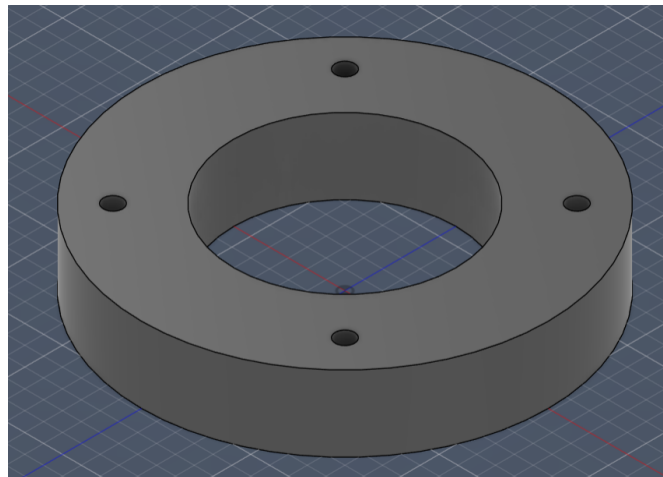


Figure 3: CAD rendering of the Eye Cup Stabilizer. The thick toroidal body provides structural rigidity. The four rod holes accept the distal ends of the guide rods, while the central opening interfaces with the rubber eye cup.

4.3.1 Design Features

- **Toroidal Body** – a thick, ring-shaped structure providing substantial mass and rigidity to resist flexing or bending of the guide rods during use.
- **Four Rod Holes** – precision bores matching those in the phone adapter, accepting the distal ends of the guide rods. These may be press-fit or secured with adhesive.
- **Central Eye Cup Mount** – the inner diameter of the ring is sized to accept a standard rubber ophthalmoscopy eye cup (ocular shell). The eye cup may be friction-fit or adhesive-bonded.

- **Stepped Inner Profile** – an internal step or ledge that acts as a stop for the eye cup, ensuring consistent positioning.

4.3.2 Functional Role

The eye cup stabilizer performs two critical functions:

1. **Structural** – it anchors the far ends of the guide rods, preventing splaying, bending, or vibration during use. Together with the phone adapter, it forms the rigid framework within which the lens carriage travels.
2. **Ergonomic** – the mounted eye cup rests gently against the patient’s orbital rim, providing a comfortable and stable point of contact. This reduces patient movement, blocks ambient light from entering the optical path, and maintains a consistent working distance between the lens and the patient’s eye.

5 Non-Printed Components

In addition to the three unique 3D-printed elements (with the lens holder ring printed twice, totalling four printed parts), the assembly requires the following commercially available components:

Component	Description	Qty	Est. Cost (PLN)
Volk 20D Lens	Aspheric double-convex condensing lens; 50 mm focal length	1	~1 675.00
Linear Bearings (LM5UU)	Linear ball bearing, 5 mm ID / 10 mm OD / 15 mm L	4	$4 \times 16.80 = 67.20$
Metal Guide Rods	Stainless steel rods, 15 cm $\times$ 5 mm dia.	4	~6.87
Rubber Eye Cup	Standard ophthalmoscopy eye cup / ocular shell	1	~15–30
Smartphone	Any smartphone with a rear camera and LED flashlight	1	(existing)
3D Printing Filament (PLA)	~50–100 g of PLA filament for all four parts	–	~5–10
Total (excluding smartphone)			~1 774–1 789

Table 2: Bill of Materials (BOM) for all components. All prices are in Polish Złoty (PLN) and reflect market prices as of mid-2026.

Cost analysis. The Volk 20D condensing lens accounts for approximately **94%** of the total prototype cost. All remaining components – the 3D-printed structure, guide rods, bearings, and eye cup – together cost under 120 PLN (~25 EUR). This makes the mechanical assembly extremely affordable and easily reproducible; the optical lens is the only component with a significant price point, and it is a standard, reusable clinical instrument that retains its value over years of service.

5.1 Guide Rod Specifications

The four metal guide rods serve as the linear rail system for the lens holder. Key specifications:

- **Material:** Stainless steel (preferred for rigidity and corrosion resistance) or aluminum (lighter weight).
- **Length:** 15 cm (150 mm).
- **Diameter:** 5 mm (must match the inner diameter of the LM5UU linear bearings).
- **Surface Finish:** Smooth, ground finish to ensure low-friction bearing contact.
- **Straightness:** Rods must be straight to within 0.1 mm over their full length to prevent binding of the lens holder assembly.

5.2 Bearing Selection

The linear bearings enable smooth, precise translation of the lens holder assembly along the guide rods. Recommended bearing type:

- **Type:** LM5UU linear ball bearing (or equivalent).
- **Inner Diameter (ID):** 5 mm (to match the guide rod diameter).
- **Outer Diameter (OD):** 10 mm (to match the bearing seats in the lens holder ring tabs).
- **Length:** 15 mm.

6 Assembly Instructions

6.1 Step 1: Prepare 3D-Printed Parts

1. Print all four parts (1× phone adapter, 2× lens holder ring, 1× eye cup stabilizer) using standard PLA printing settings.
2. Remove all support material and clean up any rough edges or stringing with a craft knife or fine sandpaper.
3. Test-fit the guide rods into each rod hole. If the fit is too tight, ream the holes gently with a 5 mm drill bit. If too loose, apply a thin layer of cyanoacrylate adhesive during final assembly.

6.2 Step 2: Install Bearings in the Lens Holder Ring

1. Select one of the two lens holder rings to serve as the bearing-equipped ring.
2. Press-fit one LM5UU linear bearing (5 mm ID / 10 mm OD) into each of its four tab bores (4 bearings total).
3. Verify that each bearing slides freely on a test rod and is fully seated flush with the tab surface.

6.3 Step 3: Assemble the Lens Holder

1. Place the bearing-equipped lens holder ring on a flat surface with the clamping face up.
2. Set the Volk 20D lens into the central aperture, resting on the annular clamping face.

3. Place the second (plain) lens holder ring on top, face-to-face, enclosing the lens between the two rings.
4. Align the rod tab holes of both rings and bond them together using cyanoacrylate or epoxy adhesive. Apply adhesive along the mating surfaces of the rings, avoiding any contact with the lens optical surfaces.
5. Allow the adhesive to cure fully before handling.
6. Verify that the lens is securely held and does not rattle or shift.

6.4 Step 4: Thread the Guide Rods

1. Insert the four 5 mm guide rods through the rod holes in the phone adapter (Component 1).
2. Slide the assembled lens holder (paired rings with lens) onto the four rods, ensuring the bearings engage smoothly with the rod surfaces.
3. Insert the distal ends of the rods into the eye cup stabilizer (Component 3).
4. Verify that the phone adapter and eye cup stabilizer are parallel and that the rods are not bowed.

6.5 Step 5: Attach the Eye Cup

1. Press-fit the rubber eye cup into the central opening of the eye cup stabilizer.
2. Ensure the eye cup protrudes sufficiently from the stabilizer to comfortably contact the patient's orbital rim.

6.6 Step 6: Mount the Smartphone

1. Slide the smartphone into the phone cradle on the rear face of the phone adapter, aligning the camera island with the dedicated cutout.
2. Press the phone firmly into the cradle until it sits flush. The perpendicular support wall prevents the phone from sliding.
3. Verify that the primary camera lens is centered over the camera aperture and that the LED flash is not occluded.

7 Usage Instructions

7.1 Pre-Examination Setup

1. Ensure the examination room is dimly lit to facilitate natural pupil dilation. **Note:** This device is designed for non-mydriatic imaging – no pharmacological dilating drops are applied to the patient's eyes.
2. Turn on the smartphone camera application and enable the LED flash (torch mode or dedicated fundus photography app).
3. Slide the lens holder assembly to an initial position approximately midway along the rods.

7.2 Image Acquisition

1. Ask the patient to look straight ahead at a fixation target.

2. Gently place the eye cup against the patient's orbital rim.
3. Observe the live camera feed on the smartphone screen.
4. Slowly slide the lens holder assembly along the rods (toward or away from the patient's eye) until a clear, focused fundus image appears on the screen.
5. Capture the image using the smartphone's shutter button or a remote trigger.
6. Repeat for the contralateral eye.

7.3 Tips for Optimal Imaging

- Reduce smartphone screen brightness to avoid glare reflecting off the lens.
- Use the smartphone's manual/pro camera mode if available to control exposure, ISO, and focus.
- Clean the Volk 20D lens surfaces before each session to avoid artifacts.
- If the fundus image appears too bright or washed out, partially occlude the LED flash with translucent tape to reduce illumination intensity.

8 Safety and Maintenance

8.1 Safety Considerations

- **Light Exposure:** The smartphone LED flash emits non-coherent, broad-spectrum visible light. Exposure duration should be minimized (seconds, not minutes) to reduce patient discomfort. The device does not use laser illumination and operates well within established ocular safety limits for incoherent light sources (ICNIRP guidelines).
- **Hygiene:** The rubber eye cup must be cleaned and disinfected with isopropyl alcohol (70 %) between patients to prevent cross-contamination.
- **Structural Integrity:** Before each use, verify that all rods are securely seated and that the lens holder assembly slides smoothly without binding. Do not use the device if any 3D-printed component shows signs of cracking or deformation.

8.2 Maintenance

- Clean all optical surfaces (Volk 20D lens, smartphone camera lens) with a microfiber cloth and lens cleaning solution.
- Periodically check bearing condition; replace if excessive play or gritty rotation is noted.
- Store the device in a protective case to prevent dust accumulation and mechanical damage.
- Replace the rubber eye cup if it becomes hardened, cracked, or deformed.

9 Limitations and Future Work

9.1 Current Limitations

- **Coaxial Illumination:** The smartphone LED flash is positioned adjacent to, rather than coaxial with, the camera lens. This off-axis illumination may introduce uneven lighting and reflections in the fundus image.

- **Manual Focus:** The current design relies on manual adjustment of the lens-to-eye distance. There is no autofocus mechanism or distance indicator.
- **Phone Compatibility:** The phone adapter cradle is designed for a specific smartphone model. Adapter variants may need to be designed for other phone models with different camera island geometries.
- **Non-Mydriatic Imaging:** The device operates without pharmacological pupil dilation. While this improves patient comfort and reduces examination time, it may limit the achievable field of view in patients with small pupils.

9.2 Planned Improvements

- Design parametric phone adapters for multiple smartphone models.
- Add a graduated distance scale to the guide rods for repeatable positioning.
- Replace the adhesive bond between the lens holder rings with threaded screw fasteners, enabling tool-free opening for lens cleaning and replacement.
- Investigate a ring LED illumination module for more uniform, coaxial lighting.

10 Summary

This document has presented the complete technical design, bill of materials, assembly procedure, and usage instructions for a low-cost, smartphone-based fundus camera prototype. The device consists of three unique 3D-printed structural components – a phone adapter with an integrated smartphone cradle, two identical lens holder rings that enclose the Volk 20D condensing lens, and an eye cup stabilizer – connected by four 5 mm × 15 cm metal guide rods. Four LM5UU linear bearings (10 mm OD / 5 mm ID) in the lens holder tabs enable smooth, adjustable positioning along the rods, providing indirect ophthalmoscopy with the smartphone’s camera and LED flash serving as the imaging sensor and illumination source, respectively.

The non-mydriatic design eliminates the need for pupil-dilating drops, improving patient comfort and reducing examination time. While the Volk 20D lens constitutes the majority of the cost (~1 675 PLN), the entire mechanical assembly costs under 120 PLN, making the structural design extremely affordable and reproducible. Integration with the project’s AI-based fundus image analysis pipeline will enable automated, point-of-care glaucoma risk assessment.